EDUCATION INTERRUPTED

Understanding the Decline of Emerging Graduates in India.



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INTRODUCTION

Have you ever wondered why, despite producing the highest number of engineers in the world, India has not won a single Nobel Prize since independence? In India, academic journeys typically follow the same pattern: students are taught to copy what the teacher writes, memorize it for exams, and forget it afterward. Einstein famously said, "*Everybody is a genius, but if you judge a fish by its ability to climb a tree, it will live its whole life believing that it is stupid*." Unfortunately, this has become a harsh reality in India. Students sitting at the back of the class often grow up considering themselves failures because the system labels them as less capable. This raises an important question for our research: What is the connection between secondary education institutions and the quality of graduates in India, and is the system effectively facilitating this connection?

EXECUTIVE SUMMARY

This research examines the factors contributing to the decline in student academic performance. It evaluates the impact of curriculum design, teaching methods, teacher qualifications, and resource availability on student engagement and outcomes. The study explores the dual role of technology in facilitating learning while also affecting attention spans and focus. Socioeconomic differences, financial pressures, and broader economic conditions are analyzed for their influence on access to education and student motivation. Additionally, mental health issues, peer influences, and time management challenges are examined alongside internal factors like motivation. Using an approach with quantitative and qualitative data, this research aims to provide insights for improving student outcomes, raising academic standards, and preparing students for higher education and careers.

PROBLEM STATEMENT



India has a large population of young adults eager to launch their careers, but they face significant challenges due to the low quality of educational institutions. These institutions fail to capitalize on the potential of the young population by providing insufficient resources and networks to support their education. Outdated teaching methodologies, which have not evolved to meet contemporary standards, discourage students and fail to address their academic and practical needs. As a result, many young individuals in India are deprived of essential knowledge and soft skills, critical for building sustainable career paths.

Furthermore, as mentioned in the article written by Prof. Arun C Mehta, it is noted that India, despite having the world's second-largest higher education system with over 700 universities and 35,000 colleges, faces critical quality challenges due to insufficient funding, outdated infrastructure, faculty shortages, and low research output. Student-faculty ratios are high, research contributions lag behind global peers, and no Indian university ranks in the top 200 for research impact. Addressing these issues requires increased public funding, private investment, better facilities, international collaboration, and merit-based faculty recruitment. Political will and systemic reforms are essential to elevate Indian universities to global standards and produce competitive graduates.

RESEARCH OBJECTIVE



This study aims to investigate the factors contributing to the decline in student academic achievement in schools and colleges, a growing concern for educators, parents, and policymakers. It examines institutional aspects such as curriculum design, teaching methods, teacher qualifications, and resource adequacy (e.g., libraries, laboratories, and extracurricular programs) to understand their impact on student engagement and performance. The study also explores classroom management, school policies, teacher-student interactions, and the alignment between teaching methods and learning preferences. Additionally, it evaluates the influence of standardized testing on fostering authentic learning and critical thinking.

This study will explore student perspectives and behaviors to understand the role of internal drive, discipline, and accountability in academic achievement. Using quantitative data from academic records and surveys, the research aims to identify the factors behind declining student performance. The findings will inform evidence-based recommendations to help institutions enhance student outcomes, raise academic standards, and better prepare students for higher education and professional careers.

RESEARCH DESIGN

INVESTIGATION QUESTIONS

What are the factors that influence the quality and availability of internships at our university?

What companies are currently offering internships for students at our university?

What sectors of the industry do these companies belong to?

How well do the current internship opportunities align with student aspirations, and how can we improve them?

The first question seeks to identify the key internal and external factors that contribute to both the quality (e.g., learning opportunities, mentorship, career development) and availability (e.g., number of opportunities, variety of roles) of internships at the university.

Next, the second question seeks to identify the specific companies and organizations that are actively offering internship opportunities to students. This is fundamental in understanding the range and diversity of internship options available.

Furthermore, the third question aims to categorize the industries of the companies offering internships, giving a more clear picture of the diversity (or lack thereof) in the types of opportunities available to students.

Finally, the last question directly examines the fit between the internship opportunities available and the career goals of students. The alignment of internships with student aspirations is critical for the effectiveness of these programs in preparing students for their future careers.

We have used various quantitative and qualitative methodologies to find data and information using our investigation and measurement questions as a part of our research. The objective of this research design is to focus on the right questions and dwell into these topics to gain insights on their solutions and finally come up with recommendations.

MEASUREMENT QUESTIONS

How many students currently have internship opportunities in our university and how we can improve it.

What are the different sectors that companies are offering internships from.

How many companies are being targeted by the university to get internship opportunities.

What is the degree for alignment between student aspirations and internship opportunities.





DATA DESIGN

We have used both primary data and secondary data to design our data.

Primary Data:

- Surveys conducted among students to know more about what they want and expect from the internship program
- Observations made in real life

Secondary Data:

- Academic journals and articles
- Web sources related to education, employment and statistics related to careers and internships
- We have also used industry reports that have information about internships and career development for drawing certain conclusions and making suggestions

SAMPLE DESIGN

Reasons why sampling is important for our research: Our sample helped us accurately represent the large population of students and their responses. The particular sample we have selected help reduce bias in our research and made the research process very efficient by categorizing different samples.

The samples that were used to collect data for our research were:

- University students who are seeking internships or who have the potential to
- Faculty and Alumni

We have used various sampling methods like:
Random Sampling
Convenience Sampling

Design Tools Used :

1. Surveys
2. Questionnaires
3. Online Survey Platforms
4. Microsoft Excel (To Manage Our Data)
5. Websites
6. External and Internal Reports.

DATA COLLECTION

These questionnaires were short and focused on basic questions, conducted through online resources.
- Surveys were distributed among students and lecturers to gather feedback on teaching methodologies, lecturer performance, and extracurricular opportunities, helping to identify areas where students benefit and areas that need improvement.
- Using structured questionnaires, we assessed the quality and impact of student portfolios, aiming to find ways to help students create stronger portfolios, either through faculty support or by increasing awareness of their importance.

Observational studies were conducted in classrooms to assess teaching effectiveness and student engagement. Participation in extracurricular activities was reviewed to determine its alignment with student needs.



DATA ANALYSIS

- The tools that we used to analyze our data are: Microsoft Excel, Excel Formulas, z-test(one sample mean & one proportional mean), and finally graphs from our Google Forms survey questionnaires. These tools were essential in our data analysis and played a crucial role in helping us find insights.

- Firstly, Excel is a powerful tool for data management and analysis. It allowed us to organize our quantitative data into rows and columns, making it easier to navigate and understand.

- Furthermore, Excel formulas enhanced our precision and speed of data analysis. We used Formulas like =AVERAGE, to gain an average score for our questions which asked for a rating between one and five.

- Secondly, Z-tests, whether for one-sample means or one-proportional means, are critical statistical tests which helped us determine if a sample data point significantly deviates from a population parameter. In the case of a one-sample mean test, the z-test assessed whether the average of a sample differs significantly from the population mean. Similarly, the one-proportional mean z-test helped us determine if a sample proportion is representative of the population proportion.

- Finally, the graphs that were automatically generated through our Google Forms survey, served as a visual representation of the data, making patterns and relationships more apparent. These graphs helped us significantly, by making our quantitative data visually accessible, and our graphs highlighted certain patterns and trends which supported our hypothesis validation.



LIMITATIONS

Research Design

Broad Research Questions:
Some investigation questions (e.g., factors affecting internship quality and availability) are too broad. Therefore, we difficult to capture all relevant factors comprehensively.

Focus on Internal Factors:
The research has overlooked external factors, like economic trends or the global job market, that also influence internship opportunities.

Sample Design

Sampling Bias: The sample does not fully represent the entire student population (e.g., only focusing on students actively seeking internships, which excludes others).

Data Collection

Reliance on Secondary Data:
Outdated Data: Secondary data (e.g., industry reports or academic articles) sources were outdated or not specific to the university's context, limiting its relevance.

PRIMARY DATA FINDINGS/INSIGHTS

Findings:

From our survey, we found out many scenarios that are affecting the readiness and preparedness of emerging graduates in India. These responses showed us the level of importance that students are being given and their outlook on further studies in India.

1. Students Share:

- A student mentioned how their science lab lacked basic equipment, making it impossible to perform practical experiments required for exams.
- Another student reported being graded unfairly in language exams because their evaluator overlooked a well-constructed answer.
- A student shared how constant pressure from parents and teachers to excel in board exams led to frequent panic attacks during preparation.
- Urban students highlighted the monotony of studying outdated syllabus content, which they felt was irrelevant to their career aspirations.

2.Lack of Institutional Support

Another problem is lack of institutional support for extracurricular activities. In fact, students reported that colleges focus more on their academic performance and treat extra-curriculars as if they are secondary or optional. This leads to underfunded, mismanaged, and inconveniently scheduled clubs and events. For instance, students reported some cases where sports practices conflicted with lectures or where the playgrounds and auditoriums were not available or were very poorly maintained.

3.Participation Inequality

Some students complain that the extracurricular activities are accessible to those who have experience already or belong to certain groups, like the sports teams or cultural committees. As a result, many other students are left out of this experience. Moreover, there was an imbalance in the variety of extracurricular opportunities that the rural colleges offered to the students in comparison with the urban institutions.

Students shared experiences of missing out on extracurricular opportunities due to academic obligations. For instance, a student from a metropolitan college mentioned how their participation in a drama club had to be abandoned because of overlapping exam schedules. A student from a rural college reported that their institution lacked a playground, which discouraged any form of physical activity or sports engagement. Another student shared that they felt left out because most cultural events were dominated by a small, experienced group, making it hard for newcomers to participate.

Insights:

1.One of the major causes of dissatisfaction is the mismatch between the skills offered in college and the demand in the industry. Several graduates feel that their curricula are outdated and unprepared to provide them with the practical or technical requirements of their jobs. For instance, graduates who enter into the technology or management sector usually complain that they were not provided with hands-on experience with modern tools and practices, making it quite challenging to get into their professional roles.

2. Another significant source of stress is low starting salary and a lack of possibilities for career advancement. More graduates say that their pay is lower than what they had hoped for or provide little direction for moving into better jobs. This problem is particularly prevalent in more labor-saturated industries, for example, marketing or entry-level work in information technology, since it reduces salaries and decreases job satisfaction.

3. Graduates also raised that career guidance and placement services are inadequately provided by the college. Many colleges are working for good placement rates rather than a correct job placement according to a student's skills and interests. Some students expressed dissatisfaction when they were forced to join an unrelated job from the subject they pursued their higher education.

4. Also, lack of preparation at the workplace makes a difference. Most graduates enter a professional environment not prepared for the demands. This includes soft skills such as communication, teamwork, and adapting. Undergraduate schools that represent only academic excellence without the real-world training experience end up unpreparing the students to enter the workforce.

RECOMMENDATIONS I

1. A more standardized grading and transparent system to help bridge those gaps is recommended. Reducing bias can also be better achieved through centrally facilitated training programmed for the evaluators along with use of tools that have artificial intelligence as such as OSM.

2. The curriculum should be redesigned based on the changing requirements. Educating students through the incorporation of Python programming, basic AI understanding, or even financial awareness can make their learning more meaningful. For instance, bringing coding into the school's curriculum can be an exposure to future jobs for high school students.

3. Holistic assessment methods should be embraced by the system, with a reduction in the importance given to final exams. There should be internal assessments, project-based learning, and extracurricular activities involved in the evaluation processes. Countries like Finland have always used such approaches and yet remain successful in education.

4. To alleviate these concerns, colleges must make it a priority to have a balanced schedule that can include academics and extracurricular activities. Having flexible timetables will ensure that students have some time for non-academic activities without compromising their studies. For example, having certain afternoons of the week for extracurricular activities can give students an opportunity to explore their interests without any academic conflicts.

RECOMMENDATIONS II

5. There are other necessary investments: infrastructure and facilities in support of extracurricular activities in colleges. The sports facilities, cultural grounds, and technical equipment need to be taken care of, and of course the funding for competitions and programs. Regular holding of college-wide sports meets, cultural festivals, inter-departmental competitions will evoke participation and lively campus surroundings.

6. Colleges should adopt policies that encourage wide participation in extracurricular activities to foster inclusivity. For instance, introducing beginner-level clubs or workshops may help engage students who lack prior experience. Equal opportunities in rural colleges can also be guaranteed through resources and mentoring to help bridge the gap in extracurricular engagement between urban and rural institutions.

7. Last but not least, it is also important to sensitize the faculty and administrators to the significance of extracurricular activities. Colleges can help students to achieve holistic development if academics and extracurriculars are considered complementary. For instance, the faculty could be motivated to incorporate real-world skills such as teamwork, leadership, and problem-solving into the assessment process, which most students learn through extracurriculars.

SECONDARY DATA FINDINGS/INSIGHTS & RECOMMENDATIONS

Findings:

A. Regarding Internships-

- Employers prioritize internship experience over other qualifications.
- Many students face barriers like financial constraints and heavy course loads that prevent them from securing internships.
- Students seek internships to gain career-related experience, but many struggle to find relevant opportunities.

B. Regarding Portfolio Building-

- Students are challenged while developing a portfolio: Many fail to relate the classroom-based activity with practical experience and lack adequate tools on how to develop a portfolio that will attract employers.

C. Regarding Professor teachings and Classroom learning-

- Effectiveness of Interactive Teaching Methods: It has been shown that teacher-student interactive methods, such as discussions, Q&A sessions, and collaborative learning, significantly promote student learning and academic performance compared to traditional teacher-centred approaches. These methods encourage active participation, which leads to better comprehension and retention of the material.
- Advantages of Student-Centred Learning: Student-centered approaches such as group work or projects are more effective than teacher-centered ones.
- Higher Learning Results: The study suggests that a student who is actively engaged in learning performs better. This can either be achieved through teacher-student interaction or student-centered approaches. An active engagement would naturally motivate and focus students and consequently improve their learning outcomes.

SECONDARY DATA FINDINGS/INSIGHTS & RECOMMENDATIONS

Insights:

A. Regarding Internships-

- Paid internships are crucial for broadening access and reducing financial barriers.
- Structured internships with clear learning goals and mentorship are linked to higher satisfaction and career readiness.
- It involves cooperation between institutions, employers, and students in setting up effective internship opportunities.

B. Regarding Portfolio Building-

- Successful building of a portfolio is positively linked to better job results; students who strategically develop different portfolios of skills such as through internships, project work, and leadership stand a better chance in grabbing jobs after graduation.

C. Regarding Professor teachings and Classroom learning-

- Significant Performance Differences: Higher mean scores were realized in students who experienced active engagement and communication when teaching.

- Teacher-Centered Methods Are Less Effective: the study reveals that students taught under traditional teacher-centered methods like lecture-based with little engagement from the students scored lower. This implies that any form of passive learning or where students are less inclined to the lesson leads to poor retention and understanding.

SECONDARY DATA FINDINGS/INSIGHTS & RECOMMENDATIONS

Recommendations:

A. Regarding Internships-

- Colleges need to partner with employers of the locality to have greater access to internships.
- Internships should be embedded in academic programs. Such programs should have clear learning plans and mentorship.
- Employers must seek to implement paid internships with a quality supervision to support better learning.

B. Regarding Portfolio Building-

- Integrate real-world experience into curriculum: The universities should encourage students to participate in internships and projects that directly align with their career goals, which helps them build a stronger professional portfolio.
- Provide mentorship programs: The institutions should offer mentorship and career advising to guide the students in choosing the right experiences and developing the portfolio strategically.
- Skill diversification: As with asset allocations, students must diversify their skills in different areas, not only technical skills but also communication skills and leadership skills to maximize employment opportunities.

C. Regarding Professor teachings and Classroom learning-

- Employ more interactive learning styles such as discussions, problem-solving, and even cooperative activities among learners. As mentioned, the former is linked with better grades.
- Activate Students: As previously mentioned, even with a student-centered classroom environment, activation by such practices as group discussions, debate, project-based learning among others encourages them to be more competent at understanding material.
- Diversity in Teaching Methodology: Instead of following the traditionally teacher-centered approach, educators will diversify different teaching practices suited to the needs and requirements of the students, thus keeping them engaged by providing multiple learning styles so that they are able to perform better.

REAL LIFE INCIDENTS

Students shared experiences of missing out on extracurricular opportunities due to academic obligations. For instance, a student from a metropolitan college mentioned how their participation in a drama club had to be abandoned because of overlapping exam schedules. A student from a rural college reported that their institution lacked a playground, which discouraged any form of physical activity or sports engagement. Another student shared that they felt left out because most cultural events were dominated by a small, experienced group, making it hard for newcomers to participate.

Graduates shared several scenarios illustrating their dissatisfaction. One graduate in computer science said their degree did not have any major on essential tools, such as R, JAVA advaned or cloud computing, which were basic requirements of the job they got after graduation. One graduate had a first job as sales executive that bore no relationship to his degree in mechanical engineering, hence frustration and no career development. According to a graduate from a lower established college, the placement cell in their respective institution focused solely on placement stats by placing students in low remunerative jobs because they could do little else but accept an unsatisfactory job.



APPENDIX I

SURVEY QUESTIONS:

Administrative questions:

1. Name
2. Job description (If you are a part of faculty)
3. Course (If you are a student)

Screening question:

Have you ever been enrolled as a student or faculty in a higher education institution?

Yes No (If no, terminate survey)

Topic A: Academic Preparedness

To assess whether students perceive themselves to be adequately prepared for their academic challenges.

1.Do you feel your previous education (high school or earlier studies) prepared you well for your current academic program?

- Yes
- No
- Somewhat

2.What specific areas of your prior education do you feel were lacking in preparing you for university-level work? (Open-ended)

3.Do you believe your academic peers are entering the program less prepared than you?

- Strongly Agree
- Agree
- Neutral
- Disagree

Topic B: Teaching Quality and Faculty Support

To investigate the impact of teaching quality on student performance, from the student perspective.

1.How would you rate the overall quality of teaching in your program?

- Excellent
- Good
- Average
- Below Average
- Poor

2.Do you feel professors and lecturers provide the necessary support to help you succeed academically?

- Always
- Most of the time
- Sometimes
- Rarely
- Never

3.What could professors do differently to help improve your academic performance? (Open-ended)

2.Have you ever experienced difficulty accessing support services (tutoring, counseling) when you needed them?

- Yes
- No

3.What improvements would you suggest for institutional resources and support services? (Open-ended)

Classification questions:

What is your age?

- 15-18
- 18-24
- 24-30
- 30+

Which cities did you attend your educational institution in? (open ended)

Topic C: Student Effort and Engagement

To examine student self-perception of effort and observations of engagement.

1.How often do you complete assigned coursework and homework on time?

- Always
- Most of the time
- Sometimes
- Rarely
- Never

2.How engaged do you feel in class discussions, projects, or other academic activities?

- Very Engaged
- Engaged
- Somewhat Engaged
- Not Engaged

3.What are the primary reasons you might not be as engaged in your studies?

- Lack of interest in the subject
- Personal challenges (financial, mental health, etc.)
- Poor teaching methods
- Other (Please specify)

Topic D: Institutional Resources and Support

To investigate the availability and effectiveness of institutional resources as seen by both students and faculty.

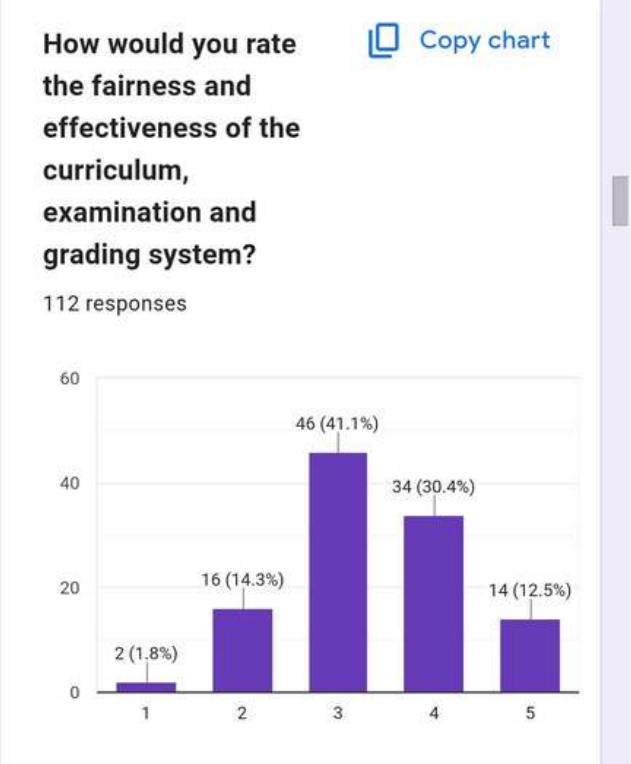
1.How accessible are academic resources (library, study materials, online tools) to support your learning?

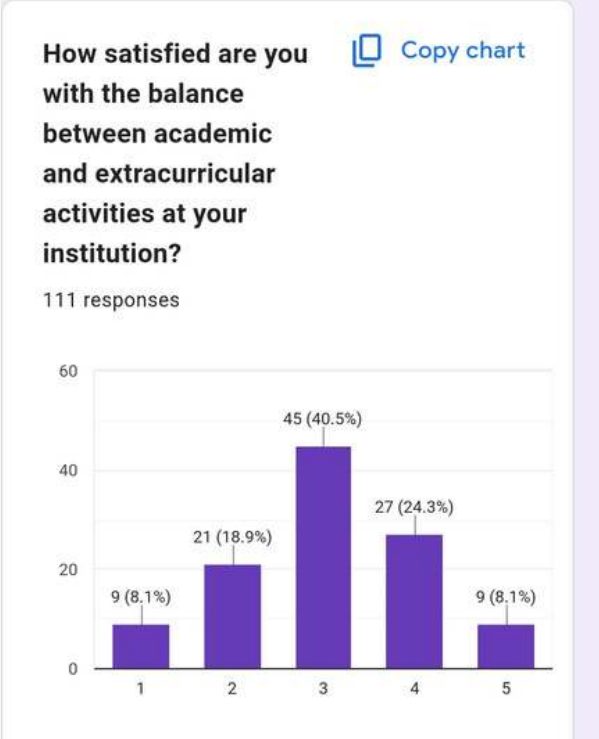
- Very Accessible
- Accessible
- Somewhat Accessible
- Not Accessible

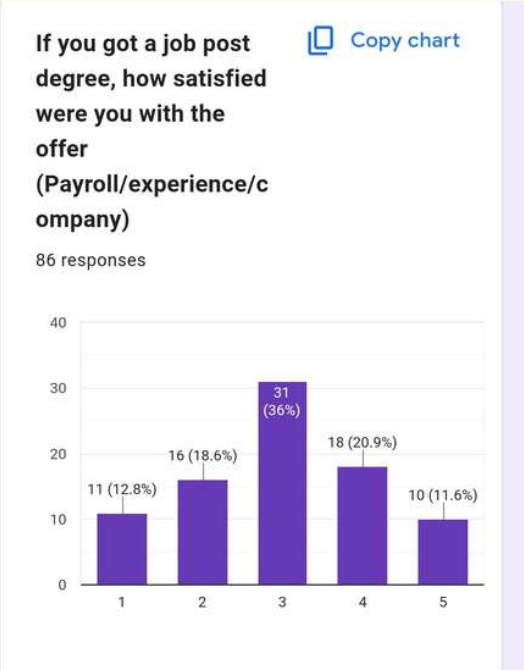
Survey Questionnaire

APPENDIX II

FEW OF THE IMPORTANT SURVEY RESPONSES



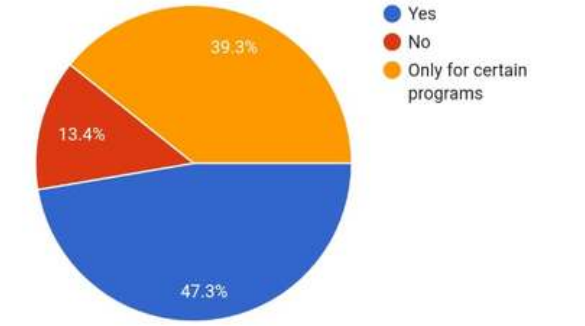




Does your curriculum require students to complete projects that can be showcased in a portfolio?

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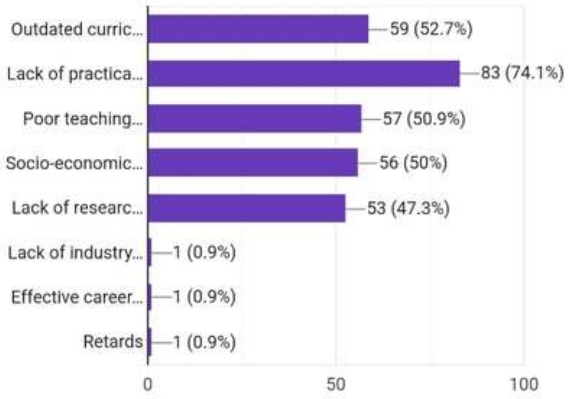
112 responses



What do you think are the key factors preventing Indian students from competing globally? (Select all that apply)

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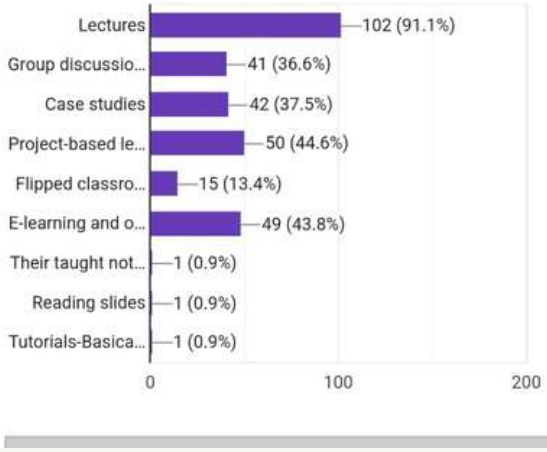
112 responses



What teaching methods are most commonly used by your teachers? (Select all that apply)

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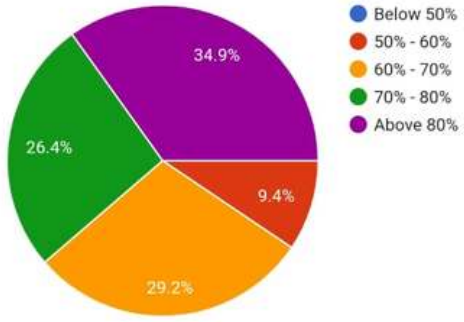
112 responses



What was your academic performance (GPA/percentage) during your degree?

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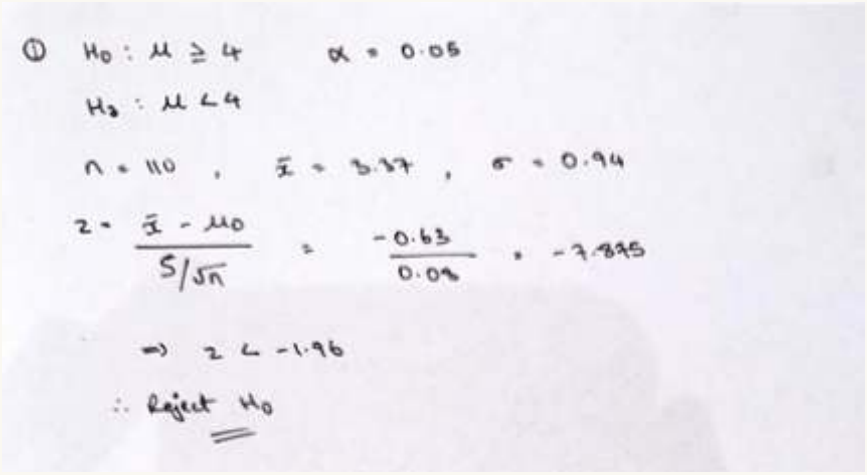
106 responses



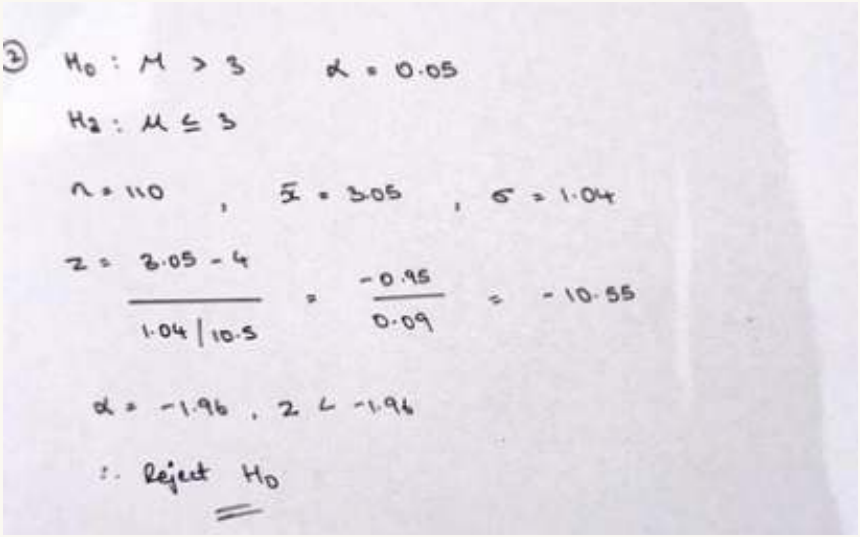
Data Collection & Analysis

APPENDIX III

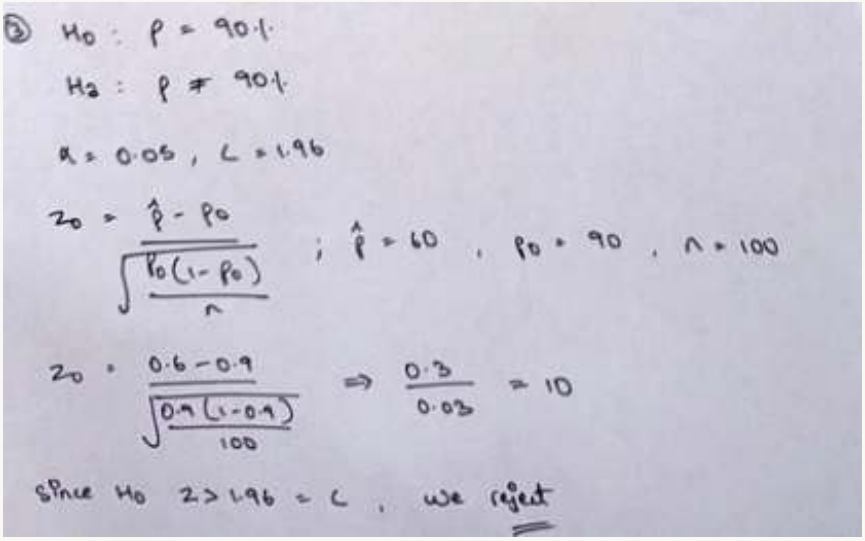
HYPOTHESIS TESTING



H0: The average rating of the fairness and effectiveness of the Indian curriculum, examination and grading system is equal to a 4 on a 5 point scale.



H0: Higher number of people are satisfied with the balance between academics and extra curricular activities in college.



H0: 90 percent of graduates are satisfied with their job post college

APPENDIX IV

https://docs.google.com/forms/d/1WmW7CdHVoYXar8zf8zxTjk0aJbv8SuzKqfj_rCy9Mlw/edit

https://docs.google.com/forms/d/13SFJn8x2Yf8zl17JjrBAERKjEE7PMLIcREtgtn5eG9w/edit

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